

ROBERTO CARRILLO AGUILAR

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EDUCATION

Worcester Polytechnic Institute

Worcester, Massachusetts

B.S. Robotics Engineering, expected May 2027 – Minor: Electrical and Computer Engineering

GPA: 3.73/4.00

Relevant Coursework: Unified Robotics IV: Navigation, Unified Robotics III: Manipulation, Real-Time Embedded Systems, Advanced Digital Systems Design with FPGAs, Unified Robotics II: Sensing, AI for Robotics, Unified Robotics I: Actuation, Embedded Systems for Robotics

PROJECTS

ROS2 Maze Exploration Robot using SLAM | *Unified Robotics IV: Navigation* | GitHub

March – May 2026

- Programmed a ROS2 robot using SLAM toolbox to map and explore a maze
- Detects frontiers, plans optimal A* paths, and avoids obstacles during navigation
- Creates map, returns to start, and computes optimal path to given goal from random pose
- Integrates SLAM, AMCL, and Nav2; tested in Gazebo simulation and on real Turtlebot

4-DoF Robotic Arm for Ball Sorting | *Unified Robotics III: Manipulation* | GitHub

August – October 2025

- Developed an autonomous sorting system implementing FK/IK/DK and trajectory planning to sort colored balls
- Implemented computer vision to detect balls and convert from camera to endeffector coordinates using transformation matrices
- Designed custom functions for real-time motion control and trajectory execution in joint- and task-space
- Combined kinematics, vision processing, and manipulation for fully automated system

Snake Game on Tiva C Board using FreeRTOS | *Real-Time Embedded Systems* | GitHub

October – December 2025

- Developed Tiva C application using FreeRTOS with tasks, queues, semaphores, and mutexes for thread-safe peripheral access
- Implemented hardware drivers for LCD, joystick, and buzzer, configuring low-level peripherals
- Integrated system monitoring to display real-time CPU, stack, and task execution metrics
- Synchronized tasks and peripherals to ensure reliable input handling and execution timing

Mock Garbage Collection Robot | *Unified Robotics II: Sensing* | GitHub

March – May 2025

- Collaborated with a team to develop an autonomous differential drive robot
- Developed gridless navigation using sensor fusion
- Incorporated an inertial sensor for precise movement control and a camera for object detection and alignment
- Utilized a load cell in conjunction with an instrumentation amplifier for trash weighing

Power Distribution and Sensor Hub PCB | *personal project* | GitHub

October – November 2025

- Designed a general-purpose power and sensor hub PCB with 3V3, 5V, and 6V outputs, as well as distance and IMU feedback
- Accounted for thermal management, ground plane isolation, and star grounding for buck converters
- Added power I/O screw terminals, as well as headers for unused MCU pins
- Partitioned power and signal ground planes, sized traces based on current, and incorporated 5V select for RGB LED array

EXPERIENCE

Technical Internship

June 2026 – July 2026

ODU Group

Mühlendorf am Inn, Germany

- Modeled and validated electropneumatic circuits in Festo FluidSim; built and tested with Festo hardware
- Assembled and diagnosed relay-based control circuits
- Machined and assembled a small spindle press from technical drawings

SKILLS

Programming Languages: C, C++, Python, MATLAB

Technical Skills: Embedded Systems, SLAM, Path Planning, Sensor Fusion, Kinematics, Computer Vision, PID Control, PCB Design

Tools: ROS2, FreeRTOS, Git, GitHub, Gazebo, KiCAD, SOLIDWORKS, Arduino

CERTIFICATES & AWARDS

Certified SOLIDWORKS Associate (Dassault Systèmes) | [Link](#)

Dean's List: four semesters [2023 - 2026] | [Link](#)

LANGUAGES

Spanish (native) | English (native) | German (B2/upper-intermediate)